Technical Design Document.

Dungeon Danger

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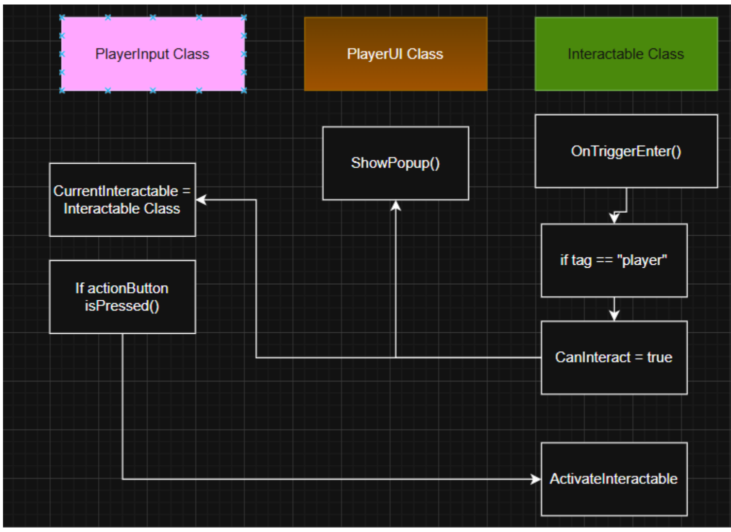
Interaction system:

The interaction system is controlled by the PlayerInteract Class and the interactable classes.

Interactable objects have a collider component set to trigger and a tag set to "Interactable".



The PlayerInteract Class uses the OnTriggerEnter( ) to check if an object tagged with "Interactable" has entered the players collider, if this is true the user interface shows that the object can be used by showing text to the player. If the player uses the action button a function will be called inside the player's InputManager class.

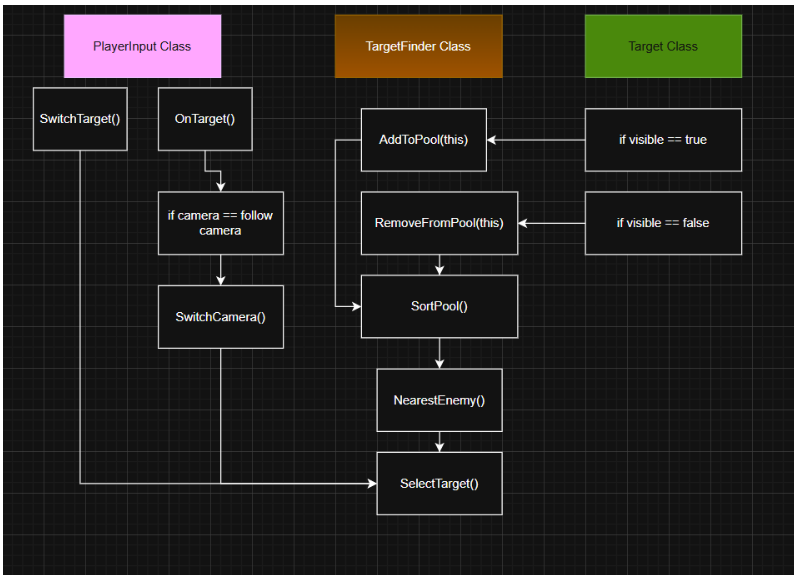
This function calls UseInteractable( ) in the PlayerInteract Class, this function checks if the player is still inside the trigger and checks if the interactable object has a class that is inherent from the Interactable base class and calls ActivateInteractable( ), if the object can only interact once the script removes itself from the interactable gameObject.

Enemy target system:

For the standard camera movement, I'm going to use Unity's Cinemachine package with the freelook camera enabled and a Cinemachine brain component added to the main camera.

The normal camera is a Cinemachine freelook camera and the enemy targeting will be done by a separate Cinamachine camera.

The enemy has a Target class that adds itself to the targeting pool if its visible inside the bounds of the camera, if its collider is not visible through the camera then the object gets deleted from the pool.



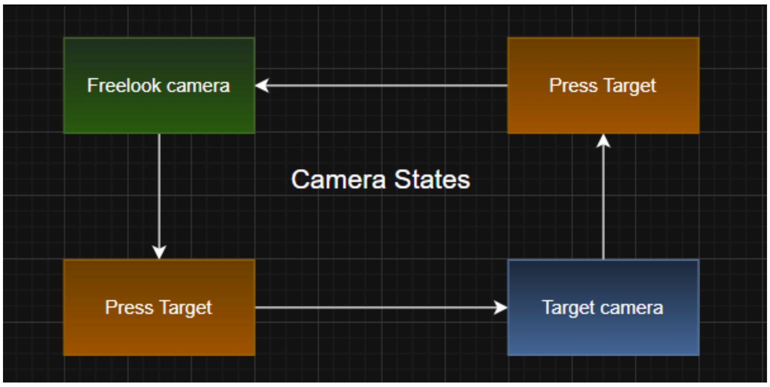
If a target is added or removed the pool gets sorted by distance from the player.

The NearestTarget ( ) loops over the pool and checks which target is closest to the center of the target camera

If the player activates the target button inside the InputManager class and the pool size is greater than 0 the camera targets the closest target by switching off the freelook camera and enabling the target camera.

If the player presses the controllers left of right shoulder buttons SwitchTarget ( ) is called and he can switch through the pools of enemies.

If the parameter is 1 then it switches to the first enemy on the right, if it's -1 than the first enemy to the left.



By pressing target again, the camera switches back to freelook mode.

If the targeted enemy dies:

If the pool size is smaller than 1, return to freelook mode.

If the pool size is greater find the closest.

Save / load system:

The save and load system will use the interaction system.

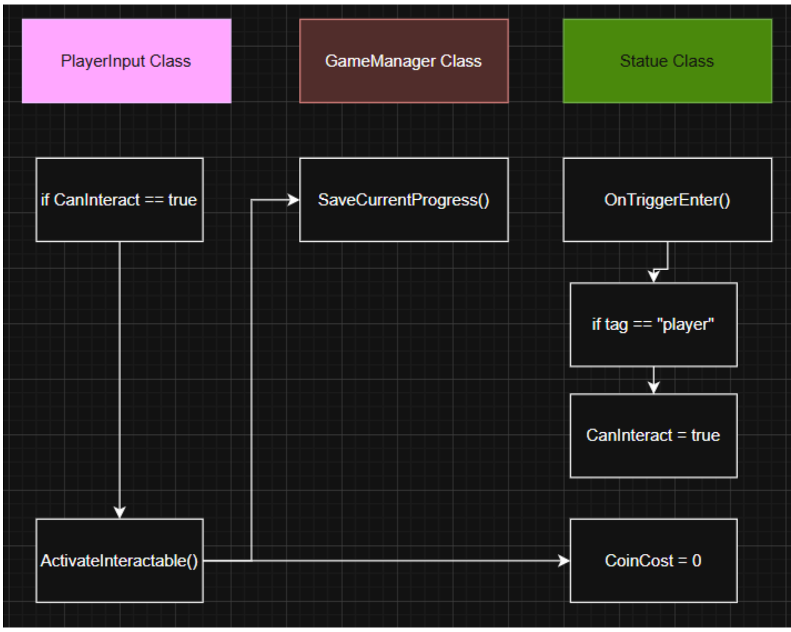
When the player comes up to the save statue a text popup shows up asking the player if he wants to buy a checkpoint for x number of coins.

If the player uses the action button ActivateInteractable ( ) checks if the player has enough coins to buy the saving point. If not then an error sound plays and nothing happens, otherwise the coin amount gets deducted from the player's total coin amount, the coin counter UI gets updated and the statues cost is set to 0 coins

(in case the player wants to use this save spot again). ActivateInteractable ( ) also send the players current resources to the GameManager ( ).

The Player Data that gets copied is:

- Position (Vector 3)
- Rotation (Vector 3)
- Health (float)
- Keys (int)
- Coins (int)



This data is saved with playerPrefs inside SaveCurrentProgress ( ).

If the player dies or restarts the game, he gets the option to load the last saved progress. If no save is available or it's the first time playing, the player is loaded with default values.

Screen Cycle & Interfaces.

The games will have the following screens:

1 StartScreen

- Logo and game title
- Start button <Ui Button> (Sends the player to the main scene)
- Quit button <Ui Button> (Closes the game)

2 MainScreen

- Player health bar <Ui Image> (Updates on damage or heal)
- Player coin counter <Ui TextMeshProGui / int > (updates on coin pickup or coin use).
- Player stamina bar <Ui Image> (Only shows when blocking)
- Key Counter <int> (updates on key use or pickup)

3 PauseScreen

- Resume button <Ui Button> (Closes the pause screen and resumes gameplay)
- Quit button <Ui button> (Opens Start scene without saving progress)

4 GameOverScreen

- Retry button <Ui button> (Load back to the latest save point)
- Quit button <Ui button> (Opens Start scene without saving progress)

5 Win / Credit Screen

- Home button <Ui button> (Opens Start scene)
- Quit button <Ui button> (Quits to desktop)